

PORTABLE AND FIXED EXTINGUISHERS & DETECTORS

Portable fire Extinguishers.

The term 'portable fire extinguishers' generally covers first-aid fire fighting appliances which can be carried by hand and from which the extinguishing agent can be expelled, usually under pressure.

They are extremely valuable for extinguishing fires at the early stages, but they cannot be used successfully to deal with large fires. They can be included in buildings having sprinklers and ~~hoses~~ hose-reels, but it is essential that staff are trained in their use.

Choice of extinguisher:

There are several factors to consider in selecting the right appliance, but the most important are as follows:

- (i) It must contain the type of extinguishing agent suitable for the fire it may be required to extinguish.
- (ii) It must not be dangerous to the user.
- (iii) It must be simple to use.
- (iv) - - - must be efficient and reliable.

Some extinguishers require more maintenance than others, and where there is a choice between extinguishers of similar type and equally suitable in all other respects, it is better to choose the type needing less maintenance.

To avoid confusion by the staff all extinguishers in the same building should, where possible, operate on the same principle, and where both water and foam extinguishers are installed they should all operate in the upright position. If an extinguisher is installed to fight a possible outbreak of fire behind cupboard or under floor boards.

it should be fitted with a short length of hose, and for rooms having unusual high ceilings the extinguisher chosen should be able to project the extinguishing agent on to the ceiling.

Some extinguishers may be too heavy or cumbersome for elderly people or women to use effectively, and therefore smaller extinguishers should be chosen for them.

High or low air temperatures (above 43°C or below 4°C) may have a detrimental effect on some types of extinguishers, and care must also be taken to choose one that will not prove dangerous or damaging in certain situations; for example, a water extinguisher should not be sited in a computer room.

Types of extinguishers

Portable fire extinguishers can be divided into the following five main groups, depending upon the extinguishing agent they contain.

Group 1: Water extinguishers

- (a) - gas pressure
- (b) - stored pressure
- (c) - Soda/acid

Group 2: Dry powder extinguishers

Group 3: Foam extinguishers

- (a) - mechanical or gas pressure
- (b) - chemical

Group 4: Carbon dioxide extinguishers

Group 5: Vapourising liquid extinguishers

- (a) - chlorobromomethane (CBM)
- (b) - bromotrifluoromethane (BTM)

- (b) Bromochlorodifluoromethane (BCF)
- (c) Bromotrifluoromethane (BTM)
- (d) Dibromotetrafluoroethane (DTE)

Water type:

Water is used on class A fires which involve fires in solid ~~combustible~~ combustible materials such as wood, paper and textile fabrics. It has better cooling properties than other extinguishing agents and is preferable for use on fires which may re-ignite if not adequately cooled.

Gas Pressure type:

The water contained in a steel cylinder is expelled by pressure from carbon dioxide gas cartridge fitted inside the extinguisher (see fig 9.1 on the handout).

To operate the ~~the~~ extinguisher the operating knob is struck, the plunger pierces a sealing disc and compressed carbon dioxide escapes from the cartridge into the space of the extinguisher body above the water level. The pressure exerted by the gas expels the water as a powerful jet through the nozzle out of the extinguisher.

Stored Pressure type

The water inside the body of the extinguisher is expelled by the pressure of either air or an inert gas such as nitrogen, which is compressed above the surface of the stored water. When the operating lever is depressed, the water is forced out of the extinguisher by the force of the compressed air or gas (see fig 9.2). The extinguisher is ready for instant use and does not require the release of pressurised carbon dioxide gas from a cartridge before

Water is expelled.

Soda/acid type:

In this type the pressure necessary for discharge of the jet of water is dependent upon the chemical reaction between bicarbonate ~~and~~ of soda dissolved in water (Sodium Sulphate) and sulphuric acid in a glass bottle. The Sulphuric acid is released into the water when the bottle is shattered, either by a metal striker (Fig 9.3) or by inverting the extinguisher (Fig 9.4). Chemical reaction takes place between the Sulphuric acid and the Sodium sulphate solution which generates Carbon dioxide gas and ^{pressurised} ~~the~~ the container, thus forcing the contents of the extinguisher through the nozzle.

Note

It is essential to keep the extinguishers the correct way up when being used, the striking type must be kept in the position shown in the dgm, while the other must be inverted. Using the extinguishers the wrong way up will allow Carbon dioxide gas to be discharged instead of the liquid.

✓ Dry Powder extinguishers;

General-purpose dry powder is suitable for all classes of fire risks and is particularly suitable for fires in inflammable liquids. The powder consists of a finely divided, non-conducting, non-toxic, water-repellent material which cools the flames, separates them from the burning materials and excludes oxygen. It also acts as a form of screen, thus enabling the operator to approach close to the fire. The dry powder stored in the body of the extinguisher is pressurised.

by nitrogen or air, which ~~expels~~ expels the powder when ~~at~~ the release from the container by when the release valve is opened (see Fig 9.5).

Alternatively, the dry powder may be expelled from the container by carbon dioxide gas after the breakage of a seal of a cartridge.

Foam Extinguishers:

Foam extinguishes fires by forming a blanket over the burning materials, thus excluding oxygen, and stays in position long enough for the hot liquid to cool. It may be used on class B fires involving burning liquids or liquefiable solids such as petrol, oil and grease.

Mechanical or gas Pressure type:

The foam compound is stored as a solution (plus F fluoroprotein) in the body of the extinguisher which is lined with polythene to prevent corrosion. The foam is expelled by an inert gas acting upon the surface of the foam (see fig 9.6)

Chemical type:

This consists of an inner ^{tube} containing chemicals dissolved in water which mix with chemical contained in the outer container when the extinguisher is inverted. Gas pressure and foam is thus generated and forced out of the nozzle by the pressure (see Fig 9.7).

Carbon dioxide Extinguishers.

Carbon dioxide gas is suitable for use on class B fires.